Development data inventory

Open Targets handoff across CRISPR screen, MAVE and Perturb-seq

Perturbation Catalogue

Development inventory

2026-10-13

Development coverage and Perturb-seq reprocessing

CRISPR screen ~1.2k datasets

MAVE ~800 datasets

Perturb-seq 30 datasets

20 of 30 Perturb-seq datasets were reprocessed from raw reads across Adamson, Norman, Nadig, Replogle and Zhu.

Study set	Datasets	DEA rows	GSEA rows
Adamson	2	1.1M	4.3K
Norman	1	1.3M	5.1K
Nadig	2	60M	0.23M
Replogle	3	170M	0.67M
Zhu	12	1.4B	6.4M
Total	20	1.7B	7.3M

Modality coverage is the current dev Elasticsearch inventory; Perturb-seq row totals are from the 20-dataset dev BigQuery snapshot queried 2026-10-13. Coverage and row counts are rounded to at most two significant digits.

Dataset metadata: study and experimental context

Common fields

- Identity: `dataset_id`, study title, year and authors
- Context: modalities, perturbation, tissue, cell type and cell line
- Scale: target and sample counts
- Full JSON also has assay, ontology, license and processing fields

Repeated annotations and counts are JSON arrays.

Example record excerpt

```
{
  "dataset_id": "adamson_2016_pilot",
  "study_title": "A Multiplexed Single-Cell CRISPR
     Screening Platform Enables Systematic Dissection of
      the Unfolded Protein Response",
  "study_year": "2016",
  "first_author": "Britt Adamson",
  "data_modalities": ["Perturb-seq"],
  "perturbation_type_labels": ["CRISPRi"],
  "cell_type_labels": ["lymphoblast"],
  "cell_line_labels": ["K 562 cell"],
  "number_of_perturbed_targets": ["8"],
  "number_of_perturbed_samples": ["5752"]
}
```

Repeated annotations and counts retain their JSON array form. `dataset_id` joins results to metadata; `perturbed_target_ensg` links to the Open Targets gene.

Perturb-seq DEA: target–effect-gene results

Column	Type	Example value: adamson_2016_pilot
perturbed_target_ensg	STRING	ENSG00000134107
perturbed_target_name	STRING	BHLHE40
effect_gene_ensg	STRING	ENSG00000094914
effect_gene_name	STRING	AAAS
log2foldchange	FLOAT64	0.0722239017
padj	FLOAT64	0.9998721657
score_name	STRING	Wilcoxon Score
score_value	FLOAT64	0.4150409698
cell_type	STRING, nullable	null

Record interpretation

The perturbed target is the intervention; the effect gene is the measured transcript. Each row reports the estimated expression change and its statistical score.

Perturb-seq GSEA: enrichment by target and gene set

Column	Type	Example value: adamson_2016_pilot
perturbed_target_ensg	STRING	ENSG00000066336
perturbed_target_symbol	STRING	SPI1
term	STRING	HALLMARK_ALLOGRAFT_REJECTION
es	FLOAT64	-0.3599006142
nes	FLOAT64	-1.1792592227
pval	FLOAT64	0.1935483871
sidak	FLOAT64	0.9999593387
fdr	FLOAT64	0.3540276454
geneset_size	INT64	81
leading_edge	ARRAY<STRING>	[SPI1, NPM1, RPS9, CTSS, RPS19, ...]
cell_type	STRING, nullable	null

Record interpretation

leading_edge is a typed string array in Parquet. In the CSV.GZ companion it is represented as a semicolon-separated list.

Per-dataset release file names

Paths are relative to the release bucket; an optional version prefix may precede `perturb-seq/`.

File	Object name pattern
Metadata	`perturb-seq/<dataset_id>.metadata.json`
DEA data	`perturb-seq/<dataset_id>.parquet`
DEA CSV mirror	`perturb-seq/<dataset_id>.csv.gz`
GSEA data	`perturb-seq/<dataset_id>.gsea.parquet`
GSEA CSV mirror	`perturb-seq/<dataset_id>.gsea.csv.gz`

- One metadata JSON plus DEA and GSEA files for each dataset ID.
- Parquet preserves typed columns; CSV.GZ uses the matching human-readable field labels.
- The metadata object contains the join key; DEA and GSEA rows are scoped by their per-dataset file names.

CRISPR screens: target-level score and interpretation

Export field	Type	Meaning
perturbed_target_ensg	STRING	Ensembl gene targeted by the screen
perturbed_target_name	STRING	Display symbol for the target
score_name	STRING	Name of the assay's score
score_value	STRING	Measured score for the target
significant	STRING	Source-provided significance call
significance_criteria	STRING	Rule or threshold used for that call

Example row: biogrid_1161

Target	Score	Value	Call / criterion
GAB3 (ENSG00000160219)	Rho (Log2e Treated vs. Untreated)	0.055800993	False / -

Interpretation

Scores are assay-specific. The significance flag travels with the source criterion so downstream users can interpret calls in the context of each screen.

MAVE: variant-level functional scores

Export field	Type	Meaning
perturbed_target_ensg	STRING	Ensembl gene associated with the variant
perturbed_target_name	STRING	Display symbol for the target
perturbation_name	STRING	Variant notation supplied by MaveDB
perturbation_position	INT64	Protein position parsed from the variant notation
perturbation_aa_wt	STRING	Wild-type amino acid parsed from the notation
perturbation_aa_change	STRING	Amino-acid change parsed from the notation
score_name	STRING	Name of the assay's score
score_value	FLOAT64	Measured functional score

Example row: urn:mavedb:00000001-a-1

Target: UBE2I (ENSG00000103275)
Variant: p.Pro73Gln (position 73)
Change: Pro → Gln
Score: exp.score = -0.0275390812623933

Interpretation

Variant identity and derived protein-change fields stay alongside the original assay score, making residue-level evidence searchable without replacing the source notation.